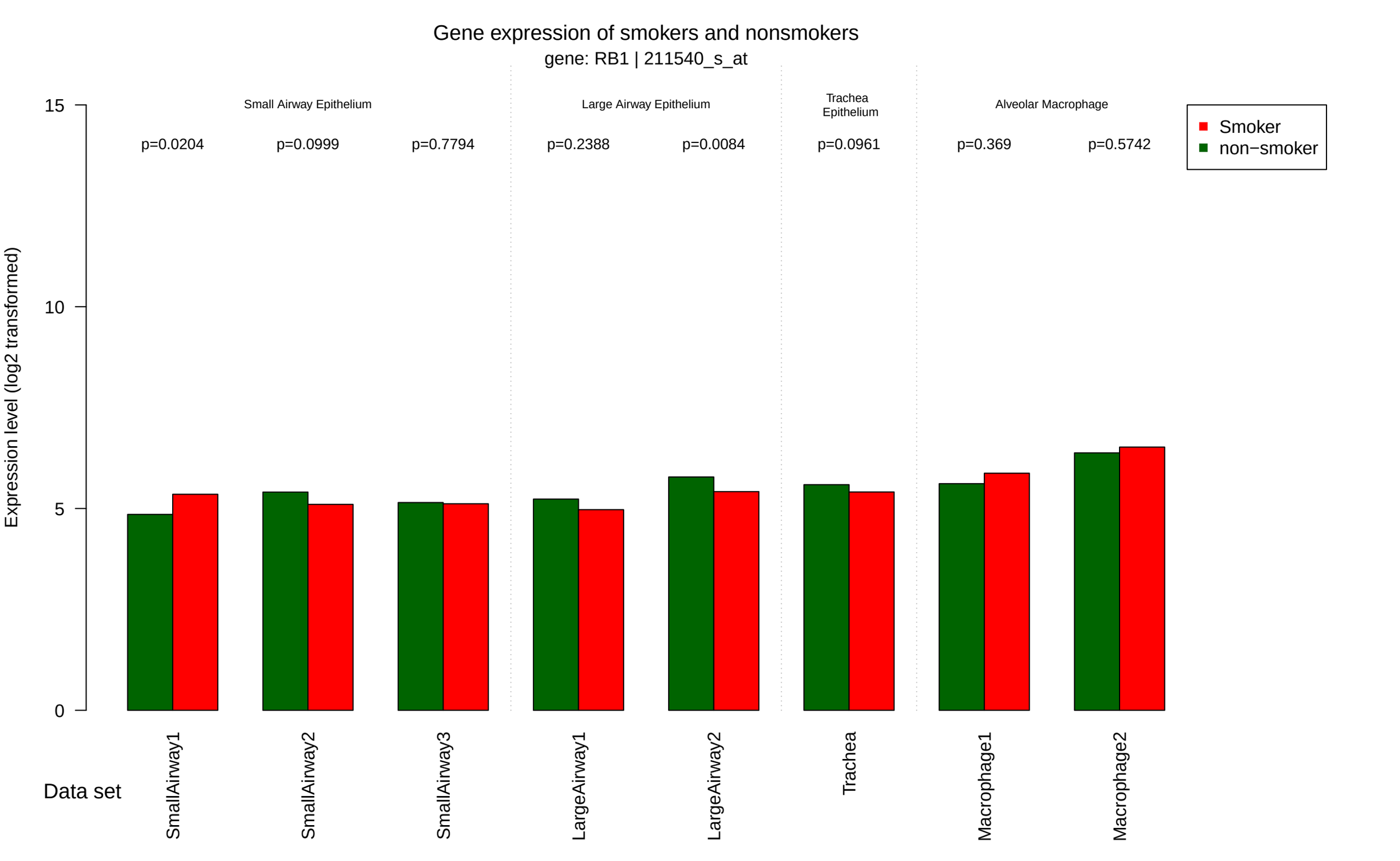
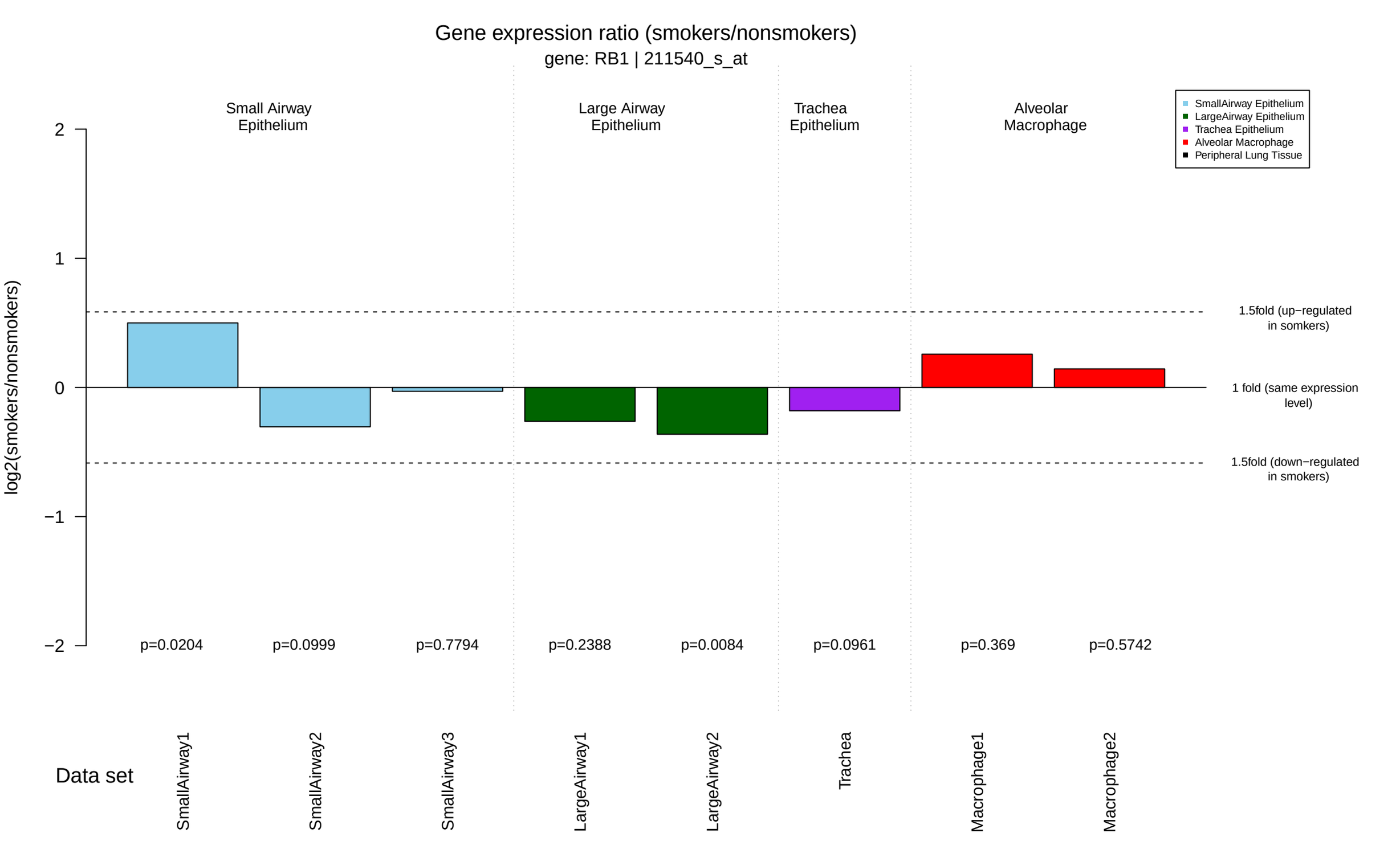
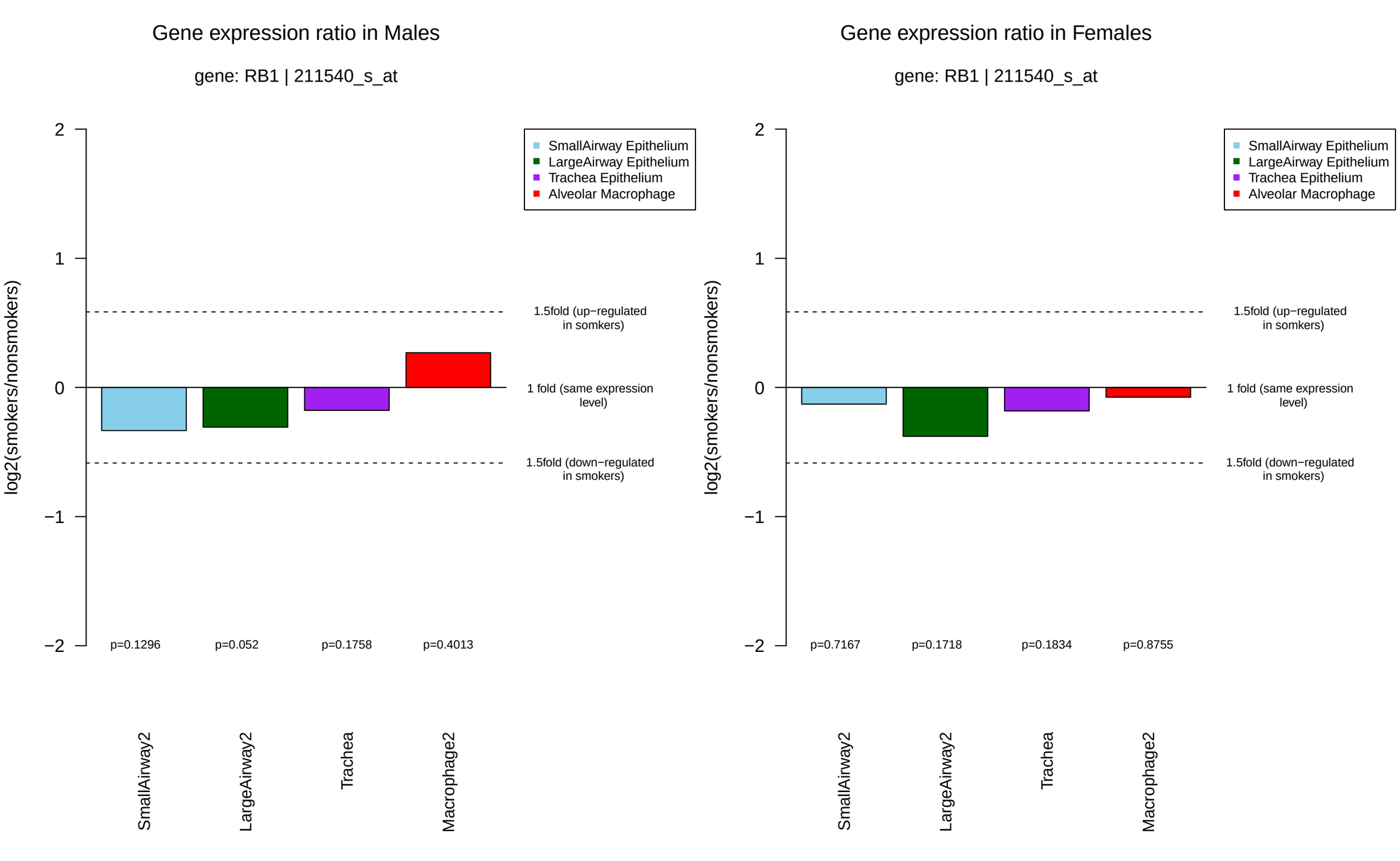
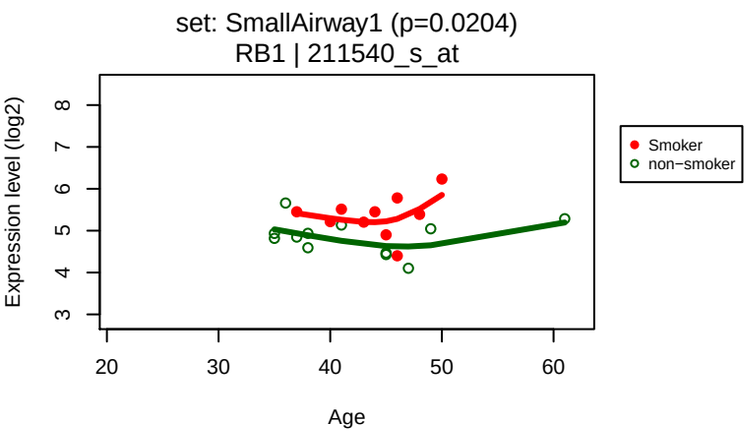
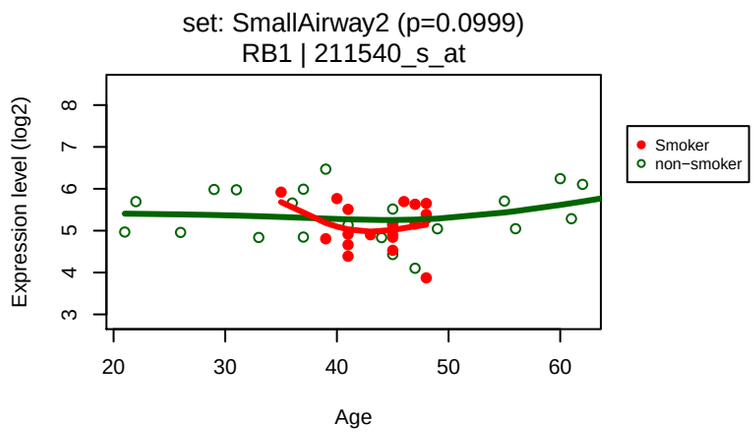
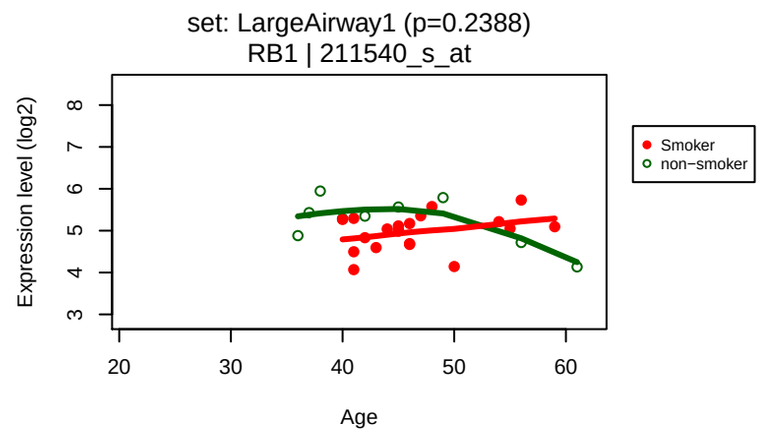

Gene expression of smokers and nonsmokers
gene: RB1 | 211540_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0204
p=0.0999
p=0.7794
p=0.2388
p=0.0084
p=0.0961
p=0.369
p=0.5742
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

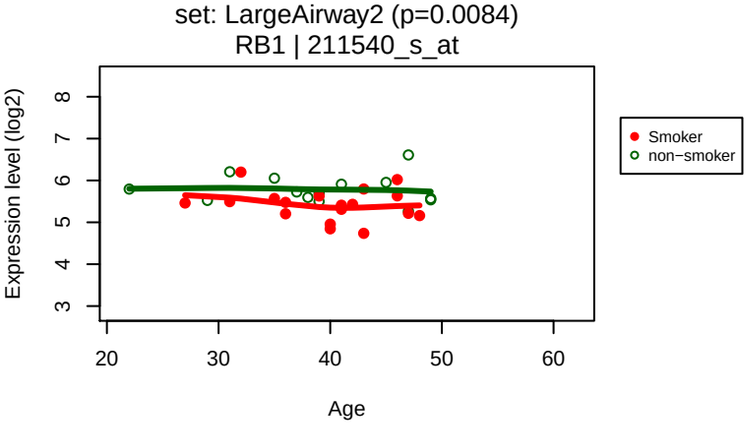

Gene expression ratio (smokers/nonsmokers)
gene: RB1 | 211540_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0204
p=0.0999
p=0.7794
p=0.2388
p=0.0084
p=0.0961
p=0.369
p=0.5742
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

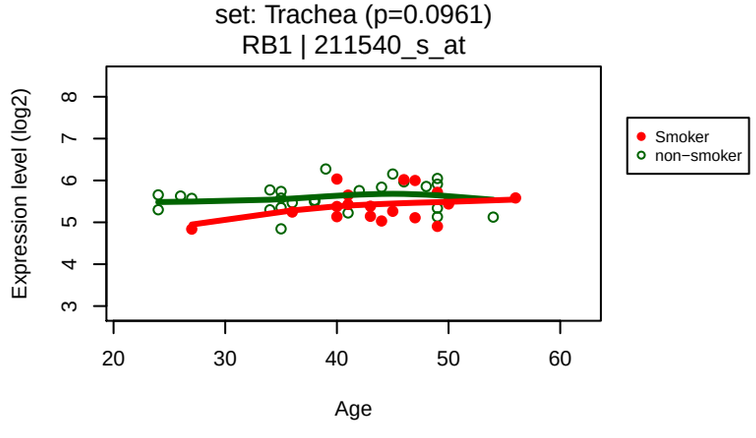

Gene expression ratio in Males
gene: RB1 | 211540_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1296
p=0.052
p=0.1758
p=0.4013
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RB1 | 211540_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7167
p=0.1718
p=0.1834
p=0.8755
SmallAirway2
LargeAirway2
Trachea
Macrophage2

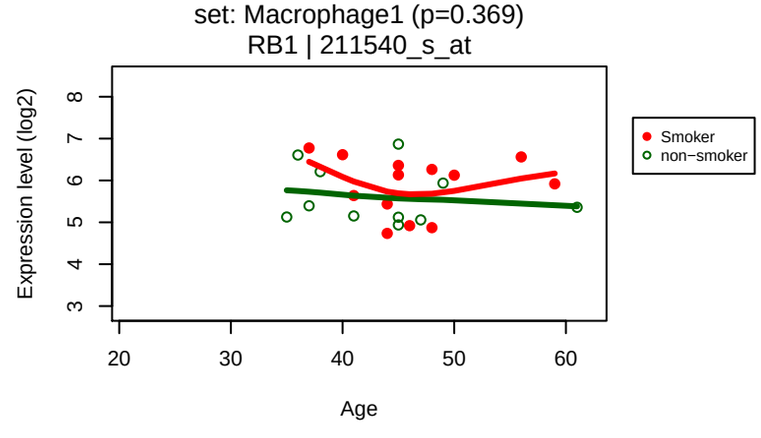

set: SmallAirway1 (p=0.0204)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker

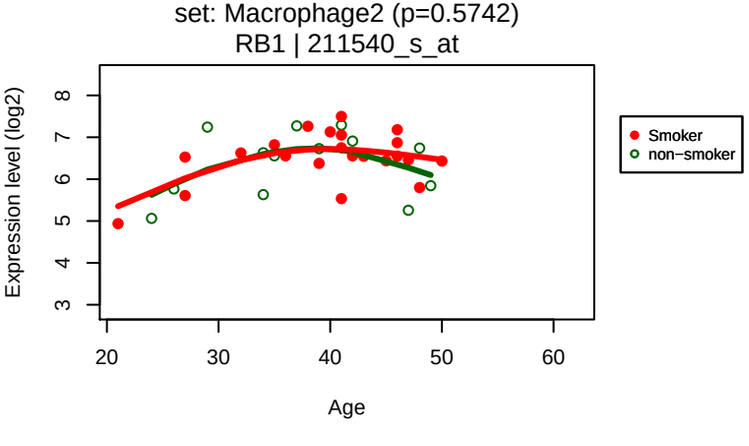

set: SmallAirway2 (p=0.0999)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker

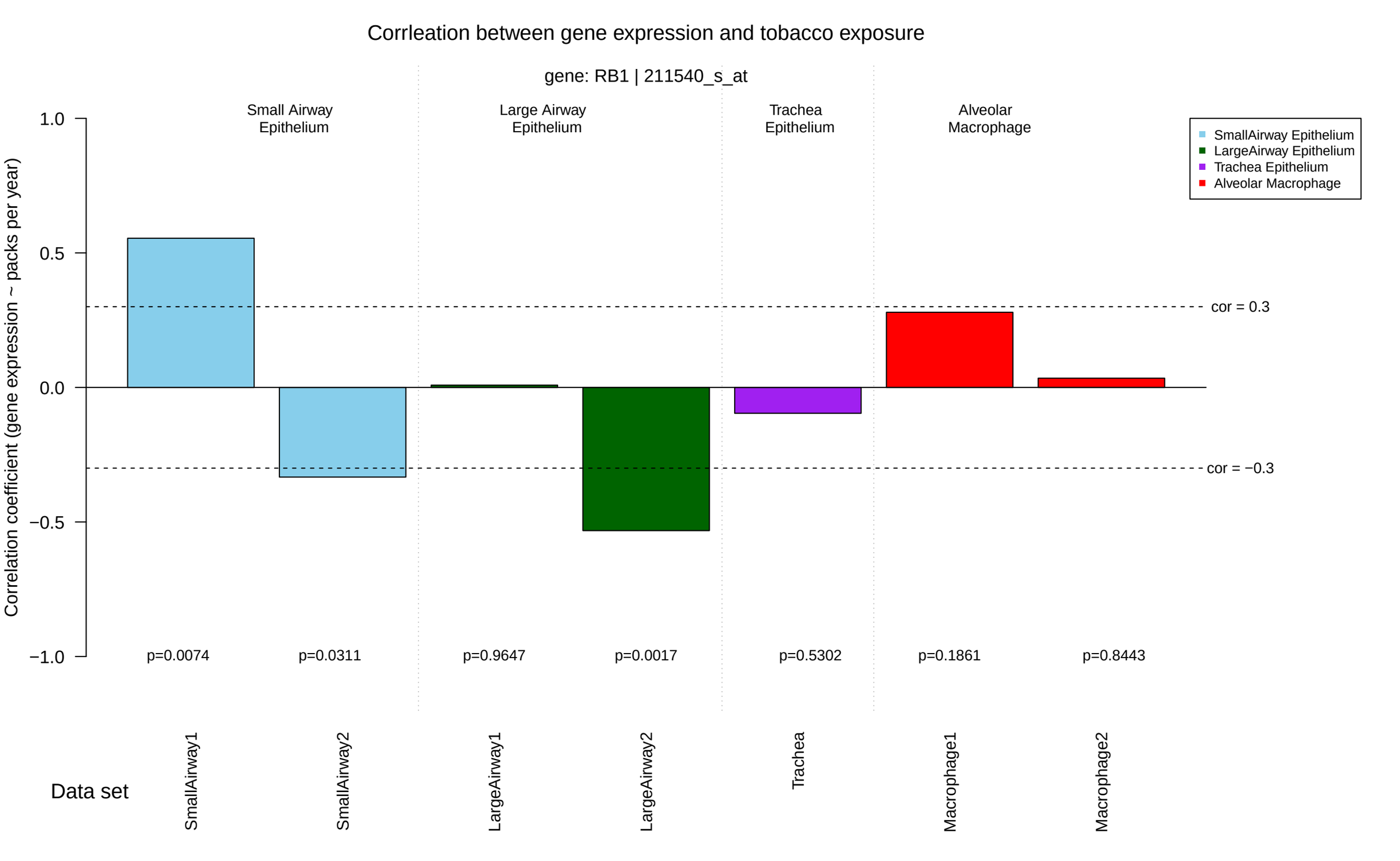

set: LargeAirway1 (p=0.2388)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0084)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0961)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.369)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5742)
RB1 | 211540_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RB1 | 211540_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0074
p=0.0311
p=0.9647
p=0.0017
p=0.5302
p=0.1861
p=0.8443
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2